

Surgeons blinded by enhanced navigation: the effect of augmented reality on attention

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Abstract

Background Advanced image-guidance systems allowing presentation of three-dimensional navigational data in real time are being developed enthusiastically for many medical procedures. Other industries, including aviation and the military, have noted that shifting attention toward such compelling assistance has detrimental effects. Using the detection rate of unexpected findings, we assess whether inattention blindness is significant in a surgical context and evaluate the impact of on-screen navigational cuing with augmented reality.

Methods Surgeons and trainees performed an endoscopic navigation exercise on a cadaveric specimen. The subjects were randomized to either a standard endoscopic view (control) or an AR view consisting of an endoscopic video fused with anatomic contours. Two unexpected findings were presented in close proximity to the target point: one

critical complication and one foreign body (screw). Task completion time, accuracy, and recognition of findings were recorded.

Results Detection of the complication was 0/15 in the AR group versus 7/17 in the control group ($p = 0.008$). Detection of the screw was 1/15 (AR) and 7/17 (control) ($p = 0.041$). Recognition of either finding was 12/17 for the control group and 1/15 for the AR group ($p < 0.001$). Accuracy was greater for the AR group than for the control group, with the median distance from the target point measuring respectively 2.10 mm (interquartile range [IQR], 1.29–2.37) and 4.13 (IQR, 3.11–7.39) ($p < 0.001$).

Conclusion Inattention blindness was evident in both groups. Although more accurate, the AR group was less likely to identify significant unexpected findings clearly within view. Advanced navigational displays may increase precision, but strategies to mitigate attentional costs need further investigation to allow safe implementation.

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Most procedural specialties have shown an increasing trend toward endoscopic, image-guided, minimally invasive techniques. Compared with open approaches, minimally invasive techniques may provide a narrower field of view and fewer spatial cues. This has resulted in a greater reliance on technology to aid in diagnosis and precise navigation.

As these procedures have evolved, there have been parallel advances in imaging technology that can provide an accurate submillimeter, three-dimensional (3D) representation of the operative field. Not surprisingly, ever

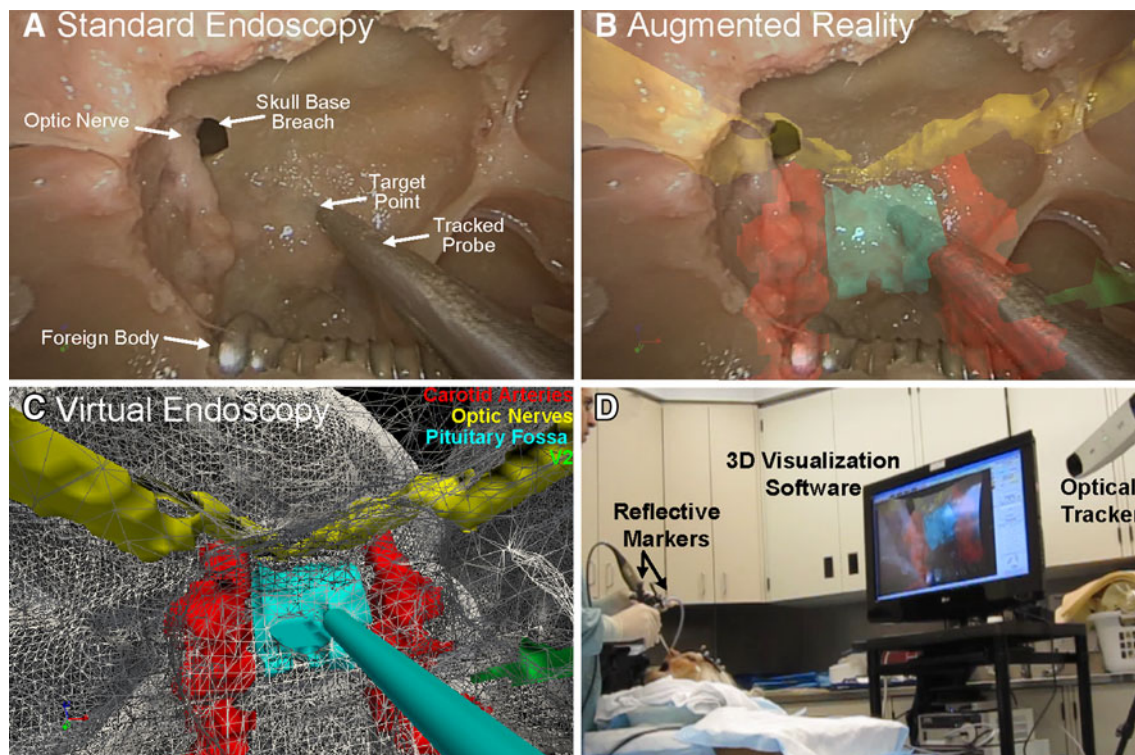


Fig. 1 **A** Still screen shot taken during the endoscopic trial of the standard view with opacity at 75 %. Relevant findings are labeled including foreign body and complication. **B** Same image as in **A** with augmented reality (AR) fusion at 25 %. This is an actual screen-shot

from a participant in the AR group who missed both findings. **C** Example showing a virtual view of contours with wire mesh surface rendering. **D** Trial setup displaying monitor and optical image-guided surgery (IGS)

increasing efforts have been made to use these data in a more dynamic way during performance of tasks.

Currently, image-guided surgery (IGS) systems are commonly used for many procedures [1], especially when the operative site is within a relatively rigid framework as in neurosurgery, skull base surgery, and sinonasal surgery. In such areas, findings have shown that IGS systems significantly change surgical strategy [2]. Technical advancements in intraoperative imaging and deformable registration [3] currently are allowing IGS to be used in areas wherein soft tissue deformation and artifact from respiratory movement and arterial pulsation had previously rendered this technology not viable [4, 5].

The Guided Therapeutics program at University Health Network, Toronto has developed a navigation system that incorporates intraoperative imaging and image registration [6, 7]. Anatomic segmentation and endoscope tracking then allows 3D representation of imaging and contouring data, which can be used during endoscopic surgery in real time [8]. This can be displayed as a virtual view on a submonitor or fused with the main endoscopic image as augmented reality (Fig. 1).

Augmented reality (AR) describes the placement of virtual objects within a real environment in real time, meaning that true and virtual images are combined in a

dynamic fashion [9]. This technique holds the promise of displaying accurate anatomic data derived from adjunct imaging on the endoscopic monitor during a procedure. Intuitively, most physicians recognize the potential clutter or visual overstimulation inherent in such a display. Although many research groups, including ours, have reported on the potential clinical benefits of this technology [8, 10–12], the possible detrimental effects of AR displays remain poorly investigated or discussed.

Human factors and psychological studies suggest that compelling displays such as AR alter attentional focus during the performance of tasks. Even in the absence of additional cues, people concentrating on a task may fail to recognize highly notable events or objects that appear within their visual field. This phenomenon, known as “inattention blindness,” becomes more evident when tasks demand greater attention [13]. It might be expected that accurate and accessible navigational data would decrease cognitive demand during a procedure and allow for a greater allocation of attention to alternative tasks [14]. However, evidence from the aviation, military, and auto industries suggests that the opposite occurs. Drawing attention to a certain cue in fact increases the chance of missing unexpected but important events occurring in the same environment [15, 16].

To refine the user interface of our image-guidance system, we aimed to assess the impact of AR on visual attention using a randomized-controlled trial comparing conventional endoscopy with an AR view (Fig. 1A, B). Endoscopic transnasal skull base navigation in a cadaveric specimen was chosen as the surgical model because it provides a complex 3D environment within a rigid framework that can be accurately registered.

Endoscopic approaches to skull base tumors are increasingly popular, and the use of image guidance in this setting is almost ubiquitous. The main outcome measure was the detection rate of two deliberately presented unexpected findings: one critical complication and one foreign body. Previous studies investigating inattention blindness presented findings that were clearly out of place, and the foreign body (a screw) was chosen to replicate these experiments in a surgical setting. We also wished to present a surgically relevant abnormality, so the critical complication (injured optic nerve) was added.

Importantly, we did not aim to assess the validity of any surgical approach or technique nor the competence of the participants. Rather, we wished to assess the effect that on-screen cuing with AR had on detection of unexpected findings as a measure of the potential attentional cost of these systems.

Methods

Otolaryngology trainees, fellows, and staff surgeons participated in the trial on a fully functional, dynamic endoscopic AR prototype after informed consent and institutional ethics approval. A single cadaver head underwent a computed tomography (CT) scan after fixation of fiducial markers. Anatomic contours of critical structures then were manually delineated using the ITK-Snap (University of Pennsylvania, PA) [17] (Fig. 1C). Sinonasal dissection, including ethmoidectomy and a wide unified sphenoidotomy, then was performed consistently with exposure for trans-sphenoidal skull base surgery such as pituitary surgery. A postablation cone-beam computed tomography (CT) image was acquired, and contours were rigidly registered to this scan [6].

An optical IGS (Polaris; NDI, Waterloo, Ontario) system was used with custom visualization software for real-time tracking of a 0° endoscope (Hopkins II telescope and IMAGE1 camera; Karl Storz, Tuttlingen, Germany) and a probe. A dynamic virtual endoscopic image then could be displayed by registering imaging and contouring data to the head [8].

Two unexpected findings were present in the specimen. First, a foreign object (small screw) was placed within the sphenoid sinus. Second, the optic nerve was divided at the

chiasm, dissected along the canal, and draped into the sinus. This represented one totally unexpected finding and one critical complication (Fig. 1A). The subjects then were randomized to perform a navigation task in one of two endoscopic environments:

1. *Standard endoscopy* (control group): The image was adjusted to 75 % opacity to match the image quality of the AR view, but no contour fusion was added (Fig. 1A).
2. *Augmented reality* (AR group): The endoscopic view was adjusted to 75 % opacity, with anatomic contours seen as a fusion in the remaining 25 % (Fig. 1B).

Each subject undertook the following task after randomization to one arm of the study. A briefing outlined the prior dissection, the AR technology, and the anatomic contours. Unaware of which view he or she would receive, each participant was asked to locate, with a tracked pointer, the anterior face of the sella turcica (pituitary fossa) in the midline as accurately and efficiently as possible. The AR group viewed precontoured anatomic data fused with the endoscopic image, which was updated dynamically in real time as the subject moved the endoscope. The subjects were aware that they were being timed, but no time limits were enforced. Randomization was stratified to balance the level of surgical experience in each group.

Once the subject indicated that he or she had located the target area, the 3D coordinates and time were recorded, and the endoscopic image was hidden. The following questions then were asked: (1) Did you notice anything unusual? (2) Did you note any critical complication? (3) Did you notice a foreign body in the sphenoid sinus? (4) Did you notice a screw? Binary responses (yes/no) were documented. Video footage of the monitor and the room were recorded separately.

Statistical analysis

Analysis was performed using SPSS Statistics 19 (Chicago, IL, USA). The recognition rates for unexpected findings between the two groups were compared using the two-sided Fisher's exact test. Spearman's rank correlations were performed to identify any significant factors associated with recognition of the findings. Binary logistic regression analysis, with Wald forward and backward modeling, was performed to assess whether randomization group, time taken, or experience level was independently associated with noting an unexpected finding. Task accuracy was assessed by measuring the distance from a "gold standard" target point set by the investigators aided by image guidance. Linear regression analysis was used to assess again the impact of group, experience level, and time on accuracy.

Results

The study enrolled 33 otolaryngology residents, fellows, and surgeons spanning all levels of experience. There was an equal distribution of experience levels between groups (likelihood ratio, 0.98). One subject in the AR group was excluded from the analysis because the AR malfunctioned due to processing latency in the laptop computer, freezing the superimposed virtual view in a static position. Retrospective scrutiny of individual video footage showed that both unexpected findings were clearly seen on the monitor in every instance.

Recognition of unexpected findings

Only 4 of the 32 subjects in the study noticed something unusual (question 1), with two commenting during the exercise.

Direct questioning about noticing a critical complication (question 2) resulted in correct answers by 7 (41 %) of 17 subjects in the control group and 0 (0 %) of 15 subjects in the AR group ($p = 0.008$). The same ratios were present for question 3 with regard to noticing a foreign body (Table 1).

One subject from the AR group altered his answer for question 4 (Did you see a screw?) from his response to question 3, meaning 1 (7 %) of 15 subjects in the AR group noticed a screw compared with 7 of 17 subjects in the control group ($p = 0.041$). Altogether, 12 subjects in the control group noticed a finding (5 noted a complication, 5 a foreign body, 2 both) compared with 1 subject in the AR group ($p < 0.001$).

The AR group showed a trend toward a shorter task completion time ($p = 0.153$) (Table 1). Logistic regression analysis showed that randomization group was the only significant independent variable for all the unexpected findings (all $p < 0.03$). After control was used for display group, time and experience level were not significant.

The median task completion time for the subgroup reporting a finding (28 s; interquartile range [IQR], 22.5–33 s) was significantly longer than for those who missed the findings (19 s; IQR, 12–30 s) ($p = 0.03$, Mann–Whitney U test). The more experienced subjects exhibited a trend toward a faster completion time ($R = 0.33$; $p = 0.064$, Spearman correlation coefficient).

Accuracy

A 3D representation of the localized points with target reference is shown in Fig. 2A. The median distance from the designated target point was significantly shorter in the AR group (2.10 mm; IQR, 1.29–2.37 mm) than in the control group (4.13 mm; IQR, 3.11–7.39 mm) ($p < 0.001$) (Table 1). Display type was the only significant variable for accuracy, but there was a trend toward greater accuracy with experience ($p = 0.074$). In Fig. 2B, localized points are graphically represented on a CT image as an anatomic reference.

Discussion

This study provides experimental evidence that surgeons exhibit inattentional blindness while performing tasks. It

Table 1 Differences in accuracy, detection of unexpected findings, and task completion time between the randomization groups

		Group A Standard endoscopy	Group B Augmented reality	<i>p</i> Value
Subjects	Total ($n = 32$)	17	15	0.98 Likelihood ratio
	PGY 1, 2 ($n = 8$)	4	4	
	PGY 3–5 ($n = 13$)	7	6	
	Fellows/surgeons ($n = 11$)	6	5	
Accuracy (mm from target)	Median (IQR)	4.13 (3.11–7.39)	2.10 (1.29–2.37)	<0.001 Mann–Whitney U test
Complication noted	Yes	7	0	0.008 Fisher’s exact test
	No	10	15	
Foreign body noted	Yes	7	0	0.008 Fisher’s exact test
	No	10	15	
Screw noted	Yes	7	1	0.04 Fisher’s exact test
	No	10	14	
Either finding noted	Yes	12	1	<0.001 Fisher’s exact test
	No	5	14	
Time to complete task (s)	Median (IQR)	25 (21–30)	19 (12–31.5)	0.15 Mann–Whitney U test

IQR interquartile range

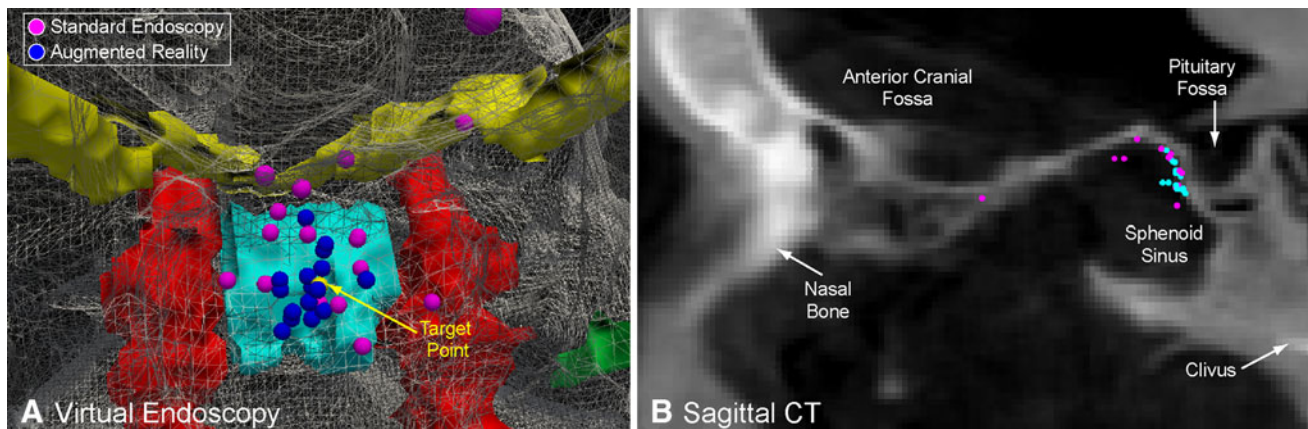


Fig. 2 **A** Three-dimensional (3D) virtual view displaying the points located by the control group (*magenta*) and the augmented reality (AR) group (*blue*). The reference target point is in yellow (partially

obscured). The AR group was more accurate ($p < 0.001$). **B** The 3D data points from **A** are shown projected onto a sagittal computed tomography (CT) image for anatomic reference

also demonstrates that recognition of significant unexpected findings can be markedly reduced when AR is used to enhance navigation.

Despite the expansion of minimally invasive techniques, image-guided procedures, and telerobotic surgery, the potential detrimental effects of advanced displays, including AR, are poorly explored in the medical literature. Some studies of anesthesia have examined the pros and cons of head-mounted AR displays for monitoring purposes, but investigation in other fields is sparse [18, 19]. Although perceptual issues with medical AR are well documented [20], we believe this is the first experiment to demonstrate that augmented surgical navigation systems have a potentially deleterious attentional cost. This has implications for the use and development of all image-guided procedures.

In our study, on-screen navigation improved accuracy, and there was a trend toward faster task completion. A trial investigating navigation on an airport surface in reduced visibility showed similar results. The addition of an augmented display resulted in more accurate navigation, faster route completion, and decreased pilot task workloads [21]. Emulating such advances in a medical context clearly is appealing because of its potential for a safer, more efficient procedure, with less demand on the physician.

Our data support other authors who suggest that such gains may be achievable for medical procedures [11, 22, 23] but that safe implementation will require some mitigation of attentional costs. The presence of inattention blindness during performance of a surgical task was expected, given the results from similar studies investigating visual perception. Even so, when we were designing this experiment, it seemed inconceivable that surgeons would miss findings so clearly visible within the operative field.

One famous example of inattention blindness is colloquially termed the “invisible gorilla.” In this example, a person in a gorilla costume is clearly present on a monitor

that the subjects are viewing while focused on a task [13]. Amazingly, during one condition, 58 % of the subjects miss the gorilla. Similarly, 59 % of our “task-focused” control arm failed to notice the screw although it would have been difficult to place it any closer to the target point (Fig. 1A).

Perhaps of greater significance, no one in the AR group noted the complication, and only one subject saw the screw. Military and aviation studies have previously documented similar drawbacks when investigating AR and head-up displays. The term “attentional tunneling” is used to describe cognitive fixation on specific cues and the resulting cost of neglecting alternative information or tasks [24]. This phenomenon has been implicated in the Three Mile Island nuclear accident [24], F16 mishaps in the U.S. Air Force [25], and cell phone-related car accidents [26, 27]. In a military simulation, detection of unexpected but “highest priority” objects was reduced when they were closely presented adjacent to cued (with AR) targets [15]. The participant’s attention was therefore drawn to the highlighted object although the noncued item nearby was of greater importance.

Attentional constraints have been shown to limit severely a pilot’s ability to process real-world and augmented display information simultaneously [28]. Head-up display symbolism may support precision landing, but attraction to the displayed image has led attention away from runway incursions [29], resulting in reduced detection of flight path objects as salient as a blimp [30].

Because we did not analyze eye movements, we are unable to determine whether failure to note the unexpected objects resulted from failure to scan the area or failure to perceive. Radiologic studies suggest that both occur and explain the “satisfaction of search” phenomenon whereby secondary abnormalities are missed on imaging studies. Greater attention, measured by gaze dwell time, is given to contrast-enhanced regions of a radiograph than to

noncontrast regions, leading to missed abnormalities [31]. This differs from other studies of eye position during radiograph interpretation, which indicated that the effect is based on failure to recognize [32]. The addition of computer-aided diagnosis in medical imaging has been shown to reduce the visual search for noncued abnormalities [33]. The differences between our randomization groups suggest that AR has an effect similar to contrast on an imaging study. Secondary findings are overlooked due to an attentional shift toward a highlighted area.

In addition to attentional tunneling, the clutter caused by on-screen cues provides another explanation for the inferior detection of unexpected findings. As more and more information is displayed within one's visual field, the costs of clutter begin to outweigh the costs of scanning [34, 35]. In a procedural context, this suggests that a point is reached at which viewing imaging data away from the operative field in a traditional manner may be safer and faster than interpreting information-rich data within the field despite the inherent shift of attention. Clutter and jitter induced by AR may reduce the user's effective visual acuity [36]. We believe there may be an additional camouflage effect, as evidenced by the difficulty seeing the optic nerve when it is fused with the red carotid artery contour (Fig. 1B).

Given our findings, we caution any claims in the medical literature that advanced navigational displays inevitably lead to safer, easier, and more precise surgery. Technical development in many disciplines has been impressive, including the introduction of live AR through continuous low-dose CT scanning, AR assistance for remote neurosurgical assistance, fused video-ultrasound images for partial nephrectomy, AR-assisted navigation during natural orifice transluminal endoscopic surgery (NOTES), and overlay AR during robotic surgery [4, 5, 11, 12, 37]. Reporting of this progress remains focused on registration accuracy, and perhaps rightly so given that it is absolutely essential for reliable navigation. However, registration error represents only one among many potential detrimental effects of AR, which include attention shifts; behavioral effects of reliance, complacency, and trust; impact of automation failure; and processing of noncued targets and cued nontargets [15, 16, 38]. In light of these other potential hazards, further investigation into interface design is required.

The sample size in the current study was relatively small, but it would be challenging to find a greater number of surgeons with varying levels of experience to participate and perform such specific and surgically relevant tasks in a realistic environment. Similar human factors studies investigating skilled end-users' use of technology have been equally limited in subjects [16, 30, 34].

A single cadaveric specimen was used. Although individual anatomic variations exist, we doubt that this would significantly affect the results.

We believe the greatest factor limiting interpretation of these results is that none of the participants had experience with operative AR. We are not sure whether lack of familiarity affected outcome. Perhaps the difference between the groups would have been insignificant if they had been given time to use the system and get used to the display.

On-screen AR could potentially be presented in various formats, and we investigated only one setting. It could be argued that an alternative view, such as a wire mesh overlay rather than solid contours or a reduced contour opacity, could limit or negate the effect demonstrated. The unknown effect of habituation on the system (learning curve) and the possibility of alternative views highlight the need for further research into display settings and feedback mechanisms.

Despite these limitations, our findings correlate closely with results in other industries. Thus, we believe the effect demonstrated does exist. Potential methods to reduce inattention blindness that require further investigation include on-demand rather than full-time AR, clutter reduction, sonifications (informative auditory feedback) [39], and a perceptually matched 3D virtual view on a submonitor.

Although the task chosen for the trial was designed to focus attention, it is important to remember that the only parameter that differed between the groups was the display setting. On-screen cuing appears to have a far greater impact on detection of unexpected findings than either experience level or time spent on the task. Despite the attentional costs, we see enormous potential benefits from real-time navigation, and we are actively engaged in addressing the issues of attention and distraction.

Our attentional capacity is limited, and we focus it for good reason; many tasks require concentration, and unexpected events are rare. Thus, quarantining a portion of our attention for unexpected events is wasteful in most circumstances and probably impossible anyway. Instead, we suggest that surgeons need to be aware of the potential detrimental effects from additional stimuli while performing procedures as advanced displays continue to be adopted into practice. Procedures and subtasks with high navigational demand should be targeted, and systems should be customized to mitigate distractive cues when demand is low. Emphasis needs to be placed on the perceptual limitations of the task-focused end user. Therefore, it is essential that physicians play a role in the development of these systems.

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